Creating Google IoT Core Project/Device for ESP32

Open the terminal window of Google IoT core and run the following commands for following operation

1. **Initialize instances**

   export PROJECT_ID="<project-id>"

   export PUBSUB_TOPIC="iotcore-topic"

   export PUBSUB_SUBSCRIPTION="iotcore-subscription"

   export REGISTRY_NAME="iotcore-registry"

   export DEVICE_NAME="esp32"

2. **Choose your region**

   export REGION_NAME="us-central1"

3. **Create a new project**

   gcloud projects create $PROJECT_ID

4. **Set default project**

   gcloud config set project $PROJECT_ID

5. **Enable Cloud Pub/Sub API**

   gcloud services enable pubsub.googleapis.com

6. **Enable Cloud IoT**

   gcloud services enable cloudiot.googleapis.com

7. **Give permission to Cloud IoT Core to publish messages on Cloud Pub/Sub**

   gcloud projects add-iam-policy-binding $PROJECT_ID --member=serviceAccount:cloud-iot@system.gserviceaccount.com --role=roles/pubsub.publisher

8. **Create Cloud Pub/Sub topic for event notifications**

   gcloud pubsub topics create $PUBSUB_TOPIC

9. **Create Cloud Pub/Sub subscription**

   gcloud pubsub subscriptions create $PUBSUB_SUBSCRIPTION --topic $PUBSUB_TOPIC

10. **Create Cloud IoT registry specifying Cloud Pub/Sub topic name**

    gcloud iot registries create $REGISTRY_NAME \

    --region=$REGION_NAME \

    --event-notification-config=topic=$PUBSUB_TOPIC \

    --enable-mqtt-config --enable-http-config

11. **Generate an Elliptic Curve (EC) ES256 private / public key pair**

    openssl ecparam -genkey -name prime256v1 -noout -out ec_private.pem

    openssl ec -in ec_private.pem -pubout -out ec_public.pem

12. **Create a new Cloud IoT device**

    gcloud iot devices create $DEVICE_NAME --region=$REGION_NAME --registry=$REGISTRY_NAME --public-key="path=./ec_public.pem,type=es256"

13. **Update the ciotc_config.h tab in the arduino code for your Wi-Fi and Cloud IoT details.**
    Set the private_key_str constant using data from the following command:

    openssl ec -in ec_private.pem -noout –text

    paste the priv key part in the ciotc_config.h tab with double colons for each line.

14. **Generate SSL certificate**
    Run the following command and paste the SSL certificate in the ciotc_config.h tab
    openssl s_client -showcerts -connect mqtt.2030.ltsapis.goog:8883

    To just create a device only carry on the process from creating a registry.